Reg. No. | | | | | | | | | |

Question Paper Code : 15302

B.E. / B.TECH. DEGREE EXAMINATION, APRIL / MAY 2019

Third Semester

B.E. – Computer Science and Engineering

15CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. What are the worst-case complexities for insertion and deletion of a key in a binary search tree?
   A) Θ (n) for both insertion and deletion
   B) Θ (log n) for insertion and deletion
   C) Θ(n) for insertion and Θ(logn) for deletion
   D) Θ(logn) for insertion and Θ(n) for deletion
2. Find out the input size and basic operation needed for searching an element in the list of n elements.
3. What is the need for ADT?
4. List some of the applications of stack.
5. The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are
   A) 64 and 5 respectively B) 63 and 6 respectively
   C) 31 and 5 respectively D) 32 and 6 respectively
6. Which of the following number of nodes can form a full binary tree?
   A) 8 B) 13 C) 14 D) 15
7. How smart union algorithm differs from simple union algorithm?
8. When will overflow and collision occur in hashing?
9. Let G be a complete undirected graph with 6 vertices. If vertices of G are labeled, then the number of distinct cycles of length 4 in G is equal to
   A) 15 B) 30 C) 45 D) 360

10. Which of the following Data structure is useful in traversing a given graph by BFS?

A) Stack B) List C) Queue D) None of the above

PART B – (5 × 16 = 80 Marks)

11. a) i. Consider the following algorithm:

ALGORITHM: Mystery(int **A, int n)

//Input: A matrix A of order n*n and order n

//Output: None

```
for (i = 0; i <n; i++)
{
    for(j = i+1; j<n; j++)
    {
        Swap(A[i][j], A[j][i])
    }
}
```

Compute Time Complexity and Space Complexity of the above algorithm and explain the intermediate steps in detail. (8 Marks)

11. a) ii. Solve the following recurrence relation for Binary Search by Backward substitution method:

$R(n) = R(n/2) + 1$ for $n > 1$, $R(1) = 1$ (8 Marks)

OR

11. b) i. Find the constants 'n' and 'c' such that t(n) is Big-Oh of g(n) where $t(n) = 1000n^2 + 50n$ and $g(n) = n^2$. (8 Marks)

11. b) ii. Write the steps for mathematical analysis of non-recursive algorithm and using that determine the order of growth for the following algorithm to find Unique elements in the array:

ALGORITHM: UniqueElements(A[0..n − 1])

//Determines whether all the elements in a given array are distinct

//Input: An array A[0..n − 1]

//Output: Returns "true" if all the elements in A are distinct

// and "false" otherwise

```
for i ←0 to n − 2 do
    for j ←i + 1 to n − 1 do
        if A[i]= A[j ]
            return false
return true
```

(8 Marks)

12. a) i. Assume that each node contains student information such as regno, name and cgpa. Demonstrate the below specified List ADT operations using Arrays with a neat diagram and devise the algorithm for the same:

A) Insert the student detail after a specified regno

B) Search and retrieve student information based on the regno (8 Marks)

12. a) ii. Evaluate the following Postfix Expression using Stack:

a b – c * d e f g / - * h i j - * + +

where a = 8, b = 3, c = 4, d = 3, e = 2, f = 7, g = 3, h = 2, i = 7 and j = 4. (8 Marks)

OR

12. b) i. Devise an algorithm to perform Polynomial Addition using List ADT implementation of linked list representation and illustrate the same with diagrammatic representation for the following:

$10X^2 + 17X + 3$

$X^5 + 6X^3 + 12X^2 + 4X + 5$ (8 Marks)

12. b) ii. Demonstrate with a neat diagram the following operations in the Circular Queue of capacity 5:

- Enqueue the elements 12, 56, 23, 11
- Dequeue 3 elements from the queue
- Now enqueue the elements 15, 17, 18
- Make the queue empty.

Note: Show the step by step diagram for each operation. (8 Marks)

13. a) i. Write down the algorithm to insert an element into the Binary Search Tree and construct binary search tree for the data given below and *show the resultant tree at each step*:

89, 12, 56, 100, 27, 10, 94, 45, 78, 15 (8 Marks)

13. a) ii. Explain how binary heap can be used for finding the kth smallest element with an example. (8 Marks)

OR

13. b) i. Build the min heap with the following elements

56, 3, 45, 21, 34, 89, 10, 33, 77, 50, 11

Show the resultant heap at each step. (8 Marks)

13. b) ii. Discuss in detail the various rotations performed in AVL tree with a neat diagram and an example. (8 Marks)

14. a) i. Explain the properties of equivalence relation and check whether <= is an equivalence relation. (8 Marks)

14. a) ii. Insert the elements 11, 44, 32, 27, 99, 54, 63, 72, 105, 88, 159, 1112, 1342, 1568 into Hash table with bucket size of 17 using Quadratic Probing technique. (8 Marks)

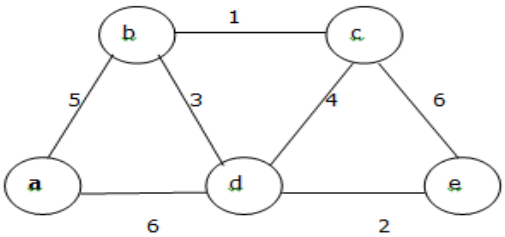
OR

14. b) i. Explain the concepts of separate chaining technique in detail and Insert the elements 131, 3, 4, 21, 61, 24, 7, 97, 8, 9 into Hash table with bucket size of 10. (8 Marks)

14. b) ii. Apply Smart Union Algorithm by Size for Union(15, 16), Union(13, 14), Union(13, 15), Union(9, 10), Union(11, 12), Union(9, 11), Union(9, 13,), Union(5, 6), Union(7, 8), Union(5, 7), Union(3, 4), Union(1, 2), Union(1, 3), Union(1, 5), Union(1, 9).

Note: Show the detailed set representation along with the array representation. Initialize the array with value as -1. (8 Marks)

15. a) i. Generate minimum spanning tree for the following graph using prim's algorithm.

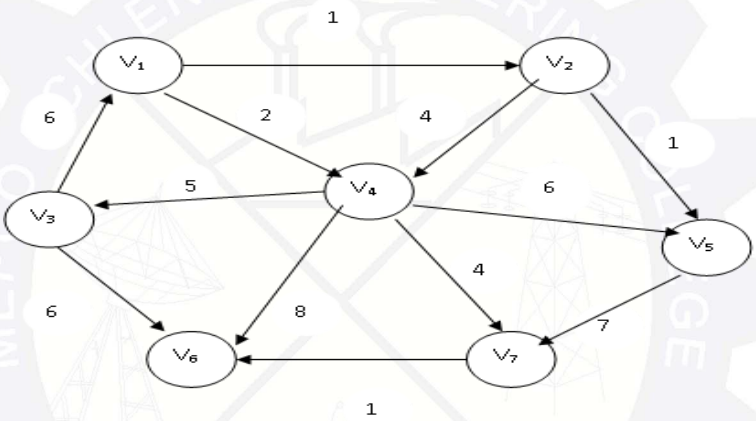


(10 Marks)

15. a) ii. Discuss in detail the different representations of a graph with appropriate examples. (6 Marks)

OR

15. b) i. For the given graph, find the shortest path with V1 as source vertex.



(10 Marks)

15. b) ii. Write a procedure for topological sorting of a graph. Give an example. (6 Marks)